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EX 5 EXPERIMENT ON

Packet Capture Tool : Wireshark

AIM: To study and analyze network packet using wireshark capture tool.

Packet Sniffer:

* Sniffs message being sent / received from / by the computer
* Store & display the contents in protocol fields.
* Passive Program:
  * never sends packet itself
  * no packets addressed to it.
  * receives a copy of all packets.

Packet sniffer Structure Diagnostic Tools:

1. Tcp dump : tcpdump -enx host 10.129.41.2 -w
2. Wireshark : Wireshark -k exe 3.out

WIRE SHARK

It is a network packet that analyzes, captures and displays network traffic in real time. It helps users monitors, analyze, troubleshoot network communication by showing packets.

Features:

* Captures live network traffic
* Decodes different network protocols
* Supports, captures & displays filters
* Provides packets, statistics and analysis

Uses

* Troubleshooting network issues
* Analysis network security
* Debugging protocol implementation.
* Learning about Network Protocol.

Capturing Packets:
After installing wireshark, open application & select a network interface to start capturing packets.

Main Panes:
* Packet List Pane: Displays Captured Packets
* Packet Details Pane: Shows detailed Protocol
* Packet Byte Pane: Displays packet data in hexadecimal format

Color Coding:
It uses different colours to identify packet types such as TCP, UDP, error pack.

Filtering Packets:
It helps to view specific packets.

Packet Inspection:
Users can select a packet to view details information follow TCP streams and analyse complete conversations between devices.

Flow Graph
Visually represent the communication between devices, making it easier to understand sequence of packet switching.

Capturing & Analysing Packets using WireShark
To filter, capture, views packets in WireShark tool.
Capture 100 packets from ethernet/IEEE 802.3 LAN interface and save it.

PROCEDURE:
1. Go to capture option
* Select stop capture automatically after 100 packets.
* Then click start capture.
* Search DNS packets in search box
* To see flow graph, click statistics flow graph.

Create a filter to display only HTTP packet & inspect packet procedure.
* Select local area connection in Wireshark
* Go to capture option.
* Select stop capture automatically after 100 packets.
* Then click start capture
* Search HTTP packets in search box
* Save the packets.
* Create filter to display only IP/ICMP.

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- Select local area connection in Wireshark
* Go to capture option.
* Select stop capture automatically after 100 packets.
* Then click start capture
* Search DHCP packets in search box.
* Save the packets

RESULT:
Network is analysed using Wireshark Capture Tool.